

The Fidelity-Utility Paradox in Surrogate-Based RL for HVAC Control

*A more accurate surrogate exposes a rougher action $\rightarrow$ temperature response, so the policy collapses on the real building;
a role-separating hybrid keeps the response smooth and the controller robust.*

Surrogate training environment

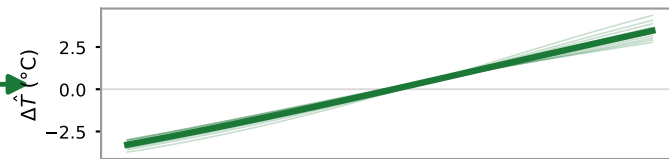
Coarse black-box v3
(1 h step — less accurate)

Accurate twin
(calibrated v3.5 / fine-res v3)

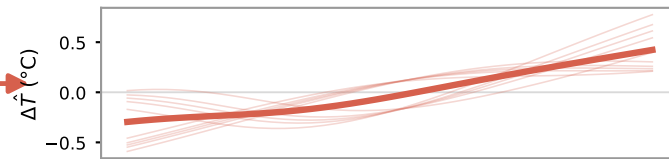
Hybrid
(v3 rollout + frozen
v3.5 censor)

Action $\rightarrow$ next-temperature response surface (measured)

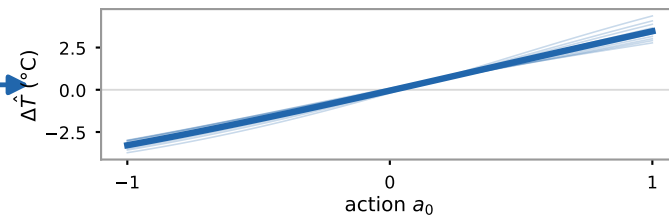
smooth · monotone



rough · non-monotone

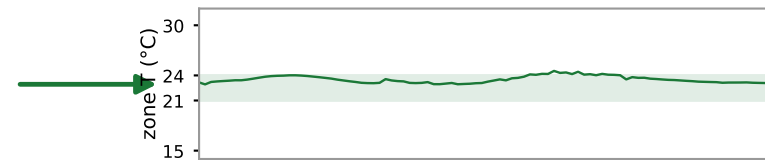


smooth (v3 dynamics)

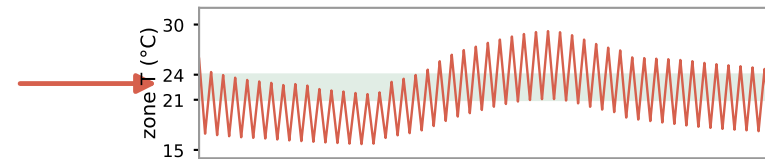


Zone temperature on the live building (measured, 24 h)

✓ **USABLE**



✗ **COLLAPSE**



✓ **ROBUST**

